

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:14

Aim

To perform image transformation using a Homography Matrix in Python with OpenCV and observe the transformed image.

Theory

A Homography matrix is a 3×3 matrix used to map points from one image plane to another.

It is commonly used for:

- Perspective transformation
- Image alignment
- Document scanning
- Image stitching
- Camera view transformation
- For four corresponding points, OpenCV can calculate the homography matrix using `cv2.findHomography()`.

Algorithm

1. Import cv2 and numpy.
2. Read the input image.
3. Define four source points from the image.
4. Define four destination points.
5. Calculate the Homography matrix using `cv2.findHomography()`.
6. Apply the Homography matrix using `cv2.warpPerspective()`.
7. Display the original and transformed images.
8. Press q to close the windows.

CODE:

```
EXP-14.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
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import cv2
import numpy as np

# Read the image
image = cv2.imread("krishna.jpeg")

if image is None:
    print("Error: Image not found!")
    exit()

# Get image dimensions
height, width = image.shape[:2]

# Source points
src_points = np.float32([
    [100, 100],
    [width - 100, 100],
    [width - 100, height - 100],
    [100, height - 100]
])

# Destination points
dst_points = np.float32([
    [0, 0],
    [width, 0],
    [width, height],
    [0, height]
])

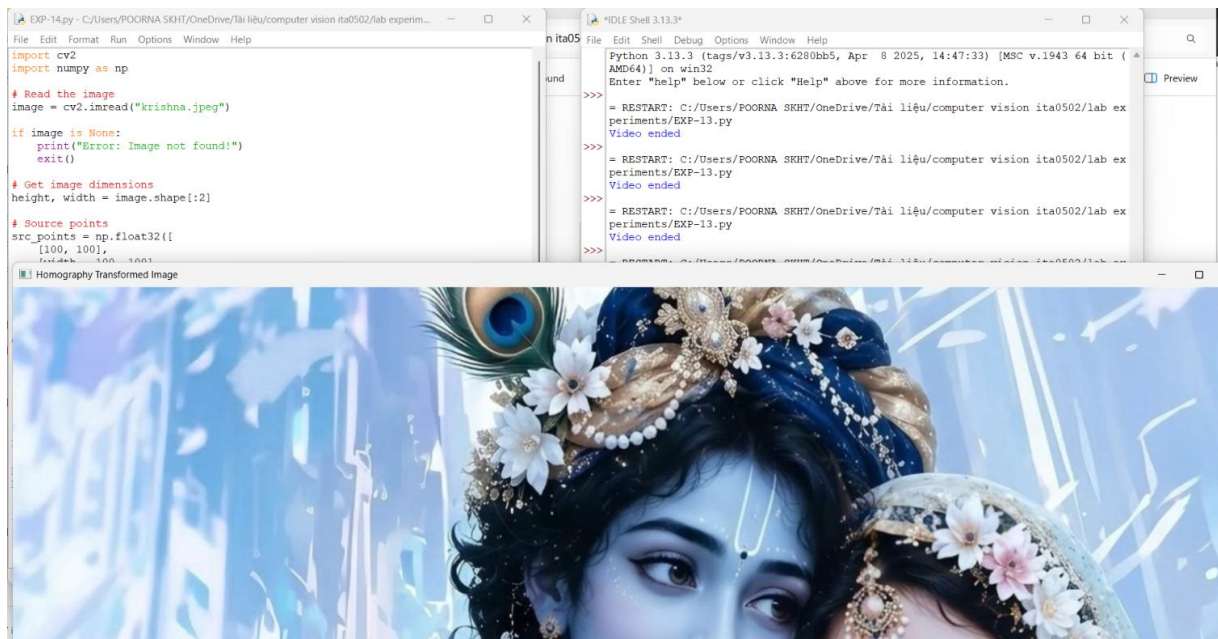
# Calculate Homography Matrix
H, status = cv2.findHomography(src_points, dst_points)

print("Homography Matrix:")
print(H)

# Apply Homography Transformation
transformed = cv2.warpPerspective(
    image, H, (width, height)
)
```

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OUTPUT :



Result:

The Homography Matrix was successfully calculated using four corresponding points, and the image was transformed using `cv2.warpPerspective()`. The perspective of the selected region was changed successfully.